

Whole Exome Sequencing Analysis

Patient name: Fetus of XX	PIN: XX
Gestational Age: 21 Weeks	Sample Number: XX
Hospital/Clinic: XX	Sample Collection Date: XX
Referring Clinician: XX	Sample Received Date: XX
Specimen: Amniotic fluid	Report Date: XX

(This test conforms to the PCPNDT act and does not determine the sex of the fetus)

Clinical History

Mrs. XX has an ongoing pregnancy. An antenatal ultrasound performed at 19 weeks' gestation revealed Stage I fetal growth restriction (estimated fetal weight <1st centile), with limited visualization of fetal anatomy due to severe oligohydramnios. Additional findings included a hypoplastic nasal bone and thoracic circumference measuring below the 1st centile. She has a significant obstetric history. Her previous male child (Baby. XXX) Presented with multiple congenital anomalies, including patent ductus arteriosus, developmental delay, congenital deafness, limb and thumb abnormalities, bilateral hip dysplasia, undescended testes, and bilateral inguinal hernia. Genetic evaluation of the affected child by whole exome sequencing with mitochondrial DNA sequencing (done elsewhere, 2023) identified a homozygous likely pathogenic variant in the *LTBP1* gene (c.1298del), a homozygous pathogenic variant in the *FANCG* gene (c.787C>T), and a heterozygous variant of uncertain significance in the *KAT6A* gene (c.4013G>A). Additionally, a heterozygous pathogenic variant in the *HBB* gene (c.92G>A), consistent with carrier status, was detected and mitochondrial DNA sequencing was normal. FISH on amniotic fluid (XX, 2026) demonstrated a normal chromosome complement for chromosomes 13, 18,21 and sex chromosomes. Conventional Karyotyping in Amniotic Fluid (XX, 2026) showed no evidence of abnormality. TORCH testing on the amniotic fluid was negative. In view of the previously identified familial pathogenic variants and the abnormal fetal ultrasound findings, Fetus of XX has been evaluated for pathogenic variations.

Results

Pathogenic variant(s) was identified in the *FANCG* gene

List of significant variant identified:

Gene	Variant*	Allele Status	Region	Disease or Disorder	Inheritance	Classification*
<i>FANCG</i> (-)	c.787C>T (p.Gln263Ter)	Homozygous	Exon 7	Fanconi anemia, complementation group G (OMIM# 614082)	Autosomal Recessive	Pathogenic (PVS1, PM2_supp, PM3, PP5)

Previously reported homozygous variant (c.787C>T; p.Gln263Ter) in *FANCG* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in homozygous state in Fetus of XX (XX, 2026) in this assay.

Variant(s) Interpretation:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
FANCG (-)	c.787C>T (p.Gln263Ter)	Autosomal Recessive	Fanconi anemia, complementation group G (OMIM# 614082)	PVS1, PM2_supp, PM3, PP5	Pathogenic
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 7	Homozygous	Stop gained	0.002% frequency observed in annotated gnomAD v4 South Asian	Ref(G): 0, Alt(A): 41, VAF: 100.00%	
In-silico Prediction Tools				Conservation Status	
LRT: -		CADD: D	MT2: D	PhyloP: C GERP++: C	
Genomic Position			Clinical & Literature Evidence [#]		
Chr9:NC_000009.12:g.35076861G>A (N7874629.2)			ClinVar: 1071538 PubMed: 30949167		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting, **Mod:** Moderate.

#Clinical and Literature Evidence: This variant has been classified as pathogenic in ClinVar database. This variant has been previously reported in patient affected with Fanconi anemia in homozygous state.

OMIM Phenotype: Fanconi anemia, complementation group G (OMIM#[614082](#)) is caused by homozygous or compound heterozygous mutation in the *FANCG* gene (OMIM#[602956](#)). Fanconi anemia (FA) is a clinically and genetically heterogeneous disorder that causes genomic instability. Characteristic clinical features include developmental abnormalities in major organ systems, early-onset bone marrow failure, and a high predisposition to cancer. The cellular hallmark of FA is hypersensitivity to DNA crosslinking agents and high frequency of chromosomal aberrations pointing to a defect in DNA repair. This disease follows autosomal recessive pattern of inheritance [2].

Previously reported homozygous variant (c.787C>T; p.Gln263Ter) in *FANCG* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in homozygous state in Fetus of XX (XX, 2026) in this assay.

Previously reported heterozygous variant of uncertain significance (c.4013G>A; p.Arg1338Lys) in *KAT6A* gene in previous child, Baby. XXX (done elsewhere, 2023) has not been detected in Fetus of XX (XX, 2026) in this assay.

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].
No other variant that warrants to be reported for the given clinical indication was identified.

List of significant carrier variant(s) identified:

Gene	Variant*	Allele Status	Region	Disease	Inheritance	Classification*
<i>LTBP1</i> [#] (+)	c.2276delC (p.Pro759GlnfsTer26)	Heterozygous	Exon 12	Cutis laxa, autosomal recessive, type IIE (OMIM# 619451)	Autosomal Recessive	Likely Pathogenic (PVS1, PM2_supp)
<i>HBB</i> (-)	c.92G>C (p.Arg31Thr)	Heterozygous	Exon 1	Beta-Thalassemia (OMIM# 613985)	Autosomal Recessive	Pathogenic (PS4, PS3_Supp, PM3_Strong, PP3, PP5)

#Previously reported homozygous variant (c.1298del) in *LTBP1* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in heterozygous state in Fetus of XX (AMB0000011850, 2026) in this assay. This variant has been reported in canonical transcript (NM_206943.4; c.2276delC; p.Pro759GlnfsTer26) in this assay.

Previously reported heterozygous variant (c.92G>C; p.Arg31Thr) in *HBB* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in heterozygous state in Fetus of XX (XX, 2026) in this assay.

Variant(s) Interpretation:

Variant 1:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
<i>LTBP1</i> (+)	c.2276delC (p.Pro759GlnfsTer26)	Autosomal Recessive	Cutis laxa, autosomal recessive, type IIE (OMIM# 619451)	PVS1, PM2_supp	Likely Pathogenic
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 12	Heterozygous	Frameshift	Not observed in annotated gnomAD v4 All	Ref(C): 18, Alt(-): 20, VAF: 52.63%	
In-silico Prediction Tools				Conservation Status	
LRT: -		CADD: D	MT2: -	PhyloP: C	GERP++: C
Genomic Position			Clinical & Literature Evidence		
Chr2:NC_000002.12:g.33257391delC (NM_206943.4)			-		

#Previously reported homozygous variant (c.1298del) in *LTBP1* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in heterozygous state in Fetus of XX (XX, 2026) in this assay. This variant has been reported in canonical transcript (NM_206943.4; c.2276delC; p.Pro759GlnfsTer26) in this assay.

Variant 2:

Gene	Variant*	Inheritance	Disease or Disorder	Criteria	Classification*
<i>HBB</i> (-)	c.92G>C (p.Arg31Thr)	Autosomal Recessive	Beta-Thalassemia (OMIM# 613985)	PS4, PS3_Supp, PM3_Strong, PP3, PP5	Pathogenic
Region	Allele Status	Sequence Ontology	Variant Frequency	Allelic Read Depths	
Exon 1	Heterozygous	Missense	0.06% frequency observed in annotated gnomAD v4 South Asian	Ref(C): 32, Alt(G): 20, VAF: 38.46%	
In-silico Prediction Tools				Conservation Status	
SIFT: D		LRT: -	PolyPhen2: D	CADD: D	MT2: D
PhyloP: C		GERP++: C			
Genomic Position			Clinical & Literature Evidence#		
Chr11:NC_000011.10:g.5226930C>G (NM_000518.5)			ClinVar: 15234 PubMed: 42181449		

Abbreviation: **D:** Deleterious; **Da:** Damaging; **Ds:** Disrupted; **C:** Conserved; **T:** Tolerated; **U:** Uncertain; **Supp:** Supporting, **Mod:** Moderate.

#Clinical and Literature Evidence: This variant has been classified as pathogenic in ClinVar database. This variant has been previously reported in patients affected with Beta-Thalassemia in compound heterozygous state.

Previously reported heterozygous variant (c.92G>C; p.Arg31Thr) in *HBB* gene in previous child, Baby. XXX (done elsewhere, 2023) has also been detected in heterozygous state in Fetus of XX (XX, 2026) in this assay.

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified.

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

MCC Status

The sample has been screened for maternal cell contamination and it is found to be negative for MCC (Refer Appendix: I).

Recommendations

- The *HBB* gene has pseudogene in human genome.
- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the Amniotic fluid was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80X to 100X using SURFSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the variant was carried out using VS- SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes

Total Read Generated	8.66 GB
Data ≥ Q30	98.03%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non-disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- The variants of uncertain significance and variations with high minor allele frequencies which are likely to be benign will be given upon request.
- In addition, this test does not cover detection of such structural variants in the genes *SMN1*, *SMN2*, *HBA1*, *HBA2*, *PMS2*, *GBA*, and *CYP21A2* which have very high homology. Moreover, the ORF15 transcript isoform (RefSeq id: NM_001034853.1) of the *RPGR* gene associated with retinitis pigmentosa is not sufficiently covered in the test.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].
- This test does not determine the sex of the fetus in adherence to the PCPNDT act.

Reference

- Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in medicine: official journal of the American College of Medical Genetics*. 17.5 (2015): 405-424.
- Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). *Nucleic Acids Res*. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.

3. Lee, Kristy et al. "ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in medicine: official journal of the American College of Medical Genetics vol. 27,8 (2025): 101454. doi:10.1016/j.gim.2025.101454.

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Appendix I

Estimation of maternal cell contamination %

No	Marker	Maternal genotype	Fetal genotype	No. of shared alleles	% MCC	Remarks
1	D3S1358	15, 17	15, 17	2	-	Non informative
2	TH01	6, 7	7	1	-	Informative
3	D21S11	31.2	31.2, 33.2	1	-	Non informative
4	D18S51	14, 17	13, 14	1	-	Informative
5	Penta E	7, 15	7, 14	1	-	Informative
6	D5S818	11, 12	11, 13	1	-	Informative
7	D13S317	10, 12	10, 12	2	-	Non informative
8	D7S820	10, 11	10	1	-	Informative
9	D16S539	12, 13	12, 13	2	-	Non informative
10	CSF1PO	9, 10	9, 13	1	-	Informative
11	Penta D	11, 12	11, 12	2	-	Non informative
12	Amelogenin	X	Not revealed	-	-	-
13	vWA	16, 19	16, 18	1	-	Informative
14	D8S1179	12, 16	14, 16	1	-	Informative
15	TPOX	9, 10	8, 10	1	-	Informative
16	FGA	21, 26	26	1	-	Informative
The average percentage of maternal cell contamination in the fetal sample is 0%						